

All random variables appearing below are assumed to be defined with respect to a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- Let $\{X_n\}_{n \in \mathbb{N}}$ be an irreducible, time-homogeneous DTMC on a finite state space $\mathcal{X}$, with TPM P and unique stationary distribution π . Given two disjoint sets $A, B \subseteq \mathcal{X}$, $A \cap B = \emptyset$, define the **probability flux** from A to B as

$$\Phi(A, B) := \sum_{x \in A} \sum_{y \in B} \pi_x P(x, y).$$

- Let $\mathcal{X}_1 \subset \mathcal{X}$ and $\mathcal{X}_2 \subset \mathcal{X}$ form a partition of $\mathcal{X}$. Show that probability flux balances, i.e.,

$$\Phi(\mathcal{X}_1, \mathcal{X}_2) = \Phi(\mathcal{X}_2, \mathcal{X}_1).$$

- Rewrite the relation $\pi = \pi P$ as a flux balance equation.

- Fix a sufficiently large number $N \in \mathbb{N}$, and let $\{X_n\}_{n=0}^N$ be an irreducible and positive recurrent DTMC with TPM P and stationary distribution π . Suppose that $X_0 \sim \pi$. For each $n \in \{0, \dots, N\}$, define $Y_n = X_{N-n}$. The process $\{Y_n\}_{n=0}^N$ is called the time-reversed chain.

- Show that $\{Y_n\}_{n=0}^N$ is a DTMC, and identify its TPM.
- What conditions should the entries of P and π satisfy for $\{X_n\}_{n=0}^N$ and $\{Y_n\}_{n=0}^N$ to have identical TPMs?

Remark: A DTMC $\{X_n\}_{n=0}^N$ which starts off in its stationary distribution is said to be **reversible** chain if its TPM is identical to the TPM of its time-reversed version.

- Let $\mathcal{G} \subset \mathcal{F}$ be a σ -algebra. For a fixed $G \in \mathcal{G}$ and $A \in \mathcal{F}$ with $\mathbb{P}(A) > 0$, evaluate

$$\frac{\int_G \mathbb{P}(A | \mathcal{G}) d\mathbb{P}}{\int_{\Omega} \mathbb{P}(A | \mathcal{G}) d\mathbb{P}}.$$

In the special case when $\mathcal{G}$ is generated by a countable partition $\{G_n\}_{n \in \mathbb{N}}$ of Ω with $\mathbb{P}(G_n) > 0$, show that one can recover Bayes' Theorem:

$$\mathbb{P}(G_i | A) = \frac{\mathbb{P}(A | G_i) \cdot \mathbb{P}(G_i)}{\sum_{j \in \mathbb{N}} \mathbb{P}(A | G_j) \cdot \mathbb{P}(G_j)}.$$

- Let X be a RV with $\mathbb{E}[X^2] < +\infty$.

- Show that for any $\mathcal{G}_1 \subset \mathcal{G}_2$,

$$\mathbb{E}\left[\left(X - \mathbb{E}[X | \mathcal{G}_1]\right)^2\right] = \mathbb{E}\left[\left(X - \mathbb{E}[X | \mathcal{G}_2]\right)^2\right] + \mathbb{E}\left[\left(\mathbb{E}[X | \mathcal{G}_2] - \mathbb{E}[X | \mathcal{G}_1]\right)^2\right].$$

- Let $\text{Var}(X | \mathcal{G}_2) := \mathbb{E}[X^2 | \mathcal{G}_2] - \left(\mathbb{E}[X | \mathcal{G}_2]\right)^2$. Using the result of part (a), show that

$$\text{Var}(X) = \mathbb{E}\left[\text{Var}(X | \mathcal{G}_2)\right] + \text{Var}\left(\mathbb{E}[X | \mathcal{G}_2]\right).$$

Hint: Think of an appropriate choice for $\mathcal{G}_1$.

- Let $Y_1, Y_2, \dots$ be IID random variables with finite mean μ and variance σ^2 , and let N be a positive integer-valued RV independent, of $\{Y_n\}_{n \in \mathbb{N}}$, with $\mathbb{E}[N^2] < +\infty$. Let $S_N := \sum_{n=1}^N Y_n$. Show that

$$\mathbb{E}[S_N] = \mu \mathbb{E}[N], \quad \text{Var}(S_N) = \sigma^2 \mathbb{E}[N] + \mu^2 \text{Var}(N).$$

6. Let $\xi_1, \xi_2, \dots$ be IID random variables with MGF $\phi(\theta)$ that is finite for all $\theta \in \mathbb{R}$. For each $n \in \mathbb{N}$, let $S_n = \sum_{k=1}^n \xi_k$. Show that

$$M_n = \frac{\exp(\theta S_n)}{(\phi(\theta))^n}, \quad n \in \mathbb{N},$$

is a martingale.

7. Let $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$ be a martingale with respect to the filtration $\mathcal{G} = \{\mathcal{G}_n\}_{n \in \mathbb{N}}$. For each $n \in \mathbb{N}$, let

$$\mathcal{H}_n := \sigma(X_1, \dots, X_n).$$

That is, $\mathcal{H} = \{\mathcal{H}_n\}_{n \in \mathbb{N}}$ denotes the natural filtration adapted to $\mathbf{X}$.

- (a) Show that $\mathcal{H}_n \subseteq \mathcal{G}_n$ for all $n \in \mathbb{N}$.
(b) Show that $\mathbf{X}$ is a martingale with respect to the filtration $\mathcal{H}$ also.
8. A process $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$ is said to be a homogeneous DTMC with TPM P and state space $\mathcal{X}$ (assumed to be finite) if

$$\mathbb{P}(\{X_{n+1} = y\} \mid \sigma(X_1, \dots, X_n)) = P(X_n, y) \quad \forall y \in \mathcal{X}, \forall n \in \mathbb{N}.$$

A function $h : \mathcal{X} \rightarrow \mathbb{R}$ is said to be P -superharmonic if

$$\forall x \in \mathcal{X}, \quad (Ph)(x) := \sum_{y \in \mathcal{X}} P(x, y) h(y) \leq h(x).$$

Show that if h is P -superharmonic, then $\{h(X_n)\}_{n \in \mathbb{N}}$ is a supermartingale with respect to the natural filtration of $\mathbf{X}$.

Note: The “super” in supermartingale comes from this superharmonic property which is associated with “ $\leq$ ” illustrated above.